20th National High School Physics Final

Solutions and Scoring Standards

•Solutions•

1.

Its radius is b. On 4 balls it rests; these 4 stay put. Each mass is m. The lower 4 centers form a square in a horizontal plane: $ABCD$ (see Fig. 20-1-1). O is where the diagonals meet. Each mutual force has size F_1.

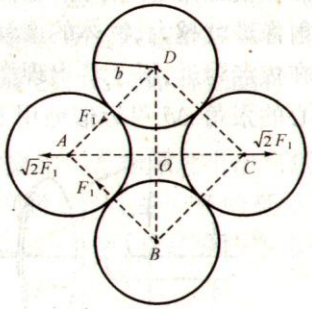


Fig. 20-1-1

All contacts are smooth. By symmetry, each ball feels a resultant $\sqrt{2}F_1$ along its diagonal.

By symmetry, diagonal AC, upper center E and bowl center O' lie in one vertical plane (Fig. 20-1-2). The upper ball exerts F_2 on each lower ball. On A, F_2 acts along AE. For the ball of radius b, we have

$$mg = 4F_2\sin\alpha \tag{1}$$

Here α is the angle of AE to horizontal AC; thus

$$\cos\alpha = \frac{AO}{AE} = \frac{\sqrt{2}a}{a+b} \tag{2}$$

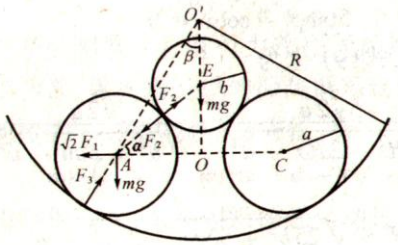


Fig. 20-1-2

Ball A feels weight mg, the resultant from B, D, namely $\sqrt{2}F_1$, upper-ball force F_2, and bowl force F_3. All pass through A by symmetry and smoothness; F_3 also passes through O'. Equilibrium gives

$$\sqrt{2}F_1 + F_2\cos\alpha - F_3\sin\beta = 0 \tag{3}$$

$$F_3\cos\beta - mg - F_2\sin\alpha = 0 \tag{4}$$

Here β is the angle of F_3 to the vertical; thus

$$\sin\beta = \frac{AO}{AO'} = \frac{\sqrt{2}a}{R-a} \tag{5}$$

From (1),

$$F_2 = \frac{mg}{4\sin\alpha} \tag{6}$$

From (4) and (6),

$$F_3 = \frac{5mg}{4\cos\beta} \tag{7}$$

From (3), (6), and (7),

$$F_1 = \frac{mg}{4\sqrt{2}}(5\tan\beta - \cot\alpha) \tag{8}$$

If $F_1 \geqslant 0$, the 4 lower balls touch; all 4 stay in place;